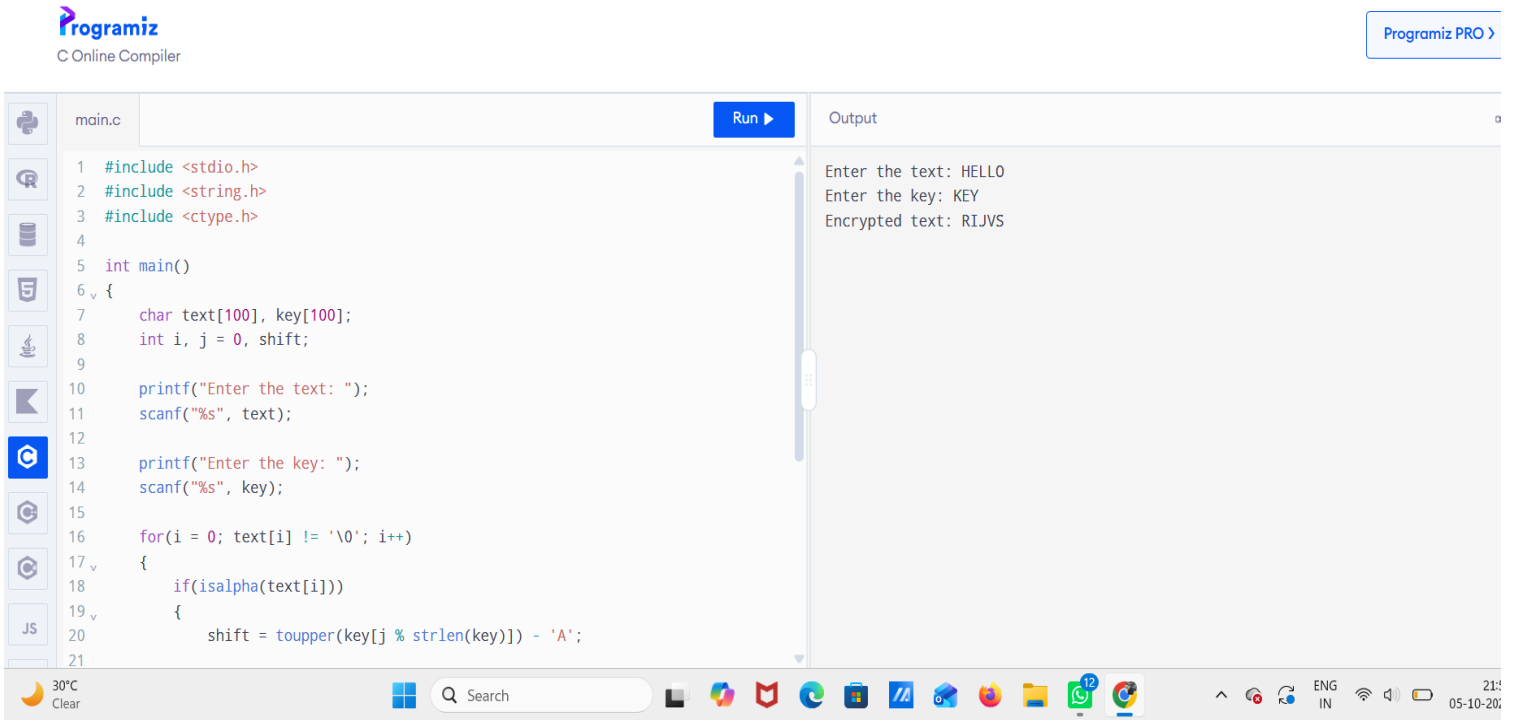
EXPERIMENT: 04

Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
   char text[100], key[100];
   int i, j = 0, shift;
   printf("Enter the text: ");
   scanf("%s", text);
   printf("Enter the key: ");
   scanf("%s", key);
   for(i = 0; text[i] != '\0'; i++)
   {
      if(isalpha(text[i]))
      {
         shift = toupper(key[j % strlen(key)]) - 'A';
         if(isupper(text[i]))
            text[i] = (text[i] - 'A' + shift) % 26 + 'A';
         else
            text[i] = (text[i] - 'a' + shift) % 26 + 'a';
         j++;
      }
   }
   printf("Encrypted text: %s\n", text);
   return 0;
}
```

OUTPUT:


Programiz
C Online Compiler
Programiz PRO
main.c
Run
Output
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
char text[100], key[100];
int i, j = 0, shift;
printf("Enter the text: ");
scanf("%s", text);
printf("Enter the key: ");
scanf("%s", key);
for(i = 0; text[i] != '\0'; i++)
{
if(isalpha(text[i]))
{
shift = toupper(key[j % strlen(key)]) - 'A';
Enter the text: HELLO
Enter the key: KEY
Encrypted text: RIJVS
30°C
Clear
Search
ENG
IN